

Smallest polyomino containing all free n -ominoes (A327094)

Minimum cells in a connected polyomino on the square grid that contains every free n -omino as a subshape under rigid motion. Values proved minimal within the search window; global minimality conjectured.

$$a(1) = 1, a(2) = 2, a(3) = 4, a(4) = 6, a(5) = 9, a(6) = 12, a(7) = 17, a(8) = 20, a(9) = 26, a(10) = 31$$

Computed by Peter Exley, May 2026

$$a(1) = 1 \text{ [PROVED] } 1 \times 1 \text{ bounding box, 1 cells.}$$



$$a(2) = 2 \text{ [PROVED] } 1 \times 2 \text{ bounding box, 2 cells.}$$



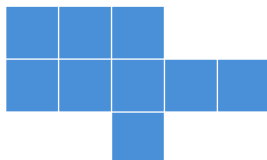
$$a(3) = 4 \text{ [PROVED] } 2 \times 3 \text{ bounding box, 4 cells.}$$



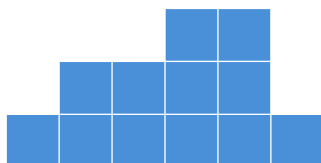
$$a(4) = 6 \text{ [PROVED] } 2 \times 4 \text{ bounding box, 6 cells.}$$



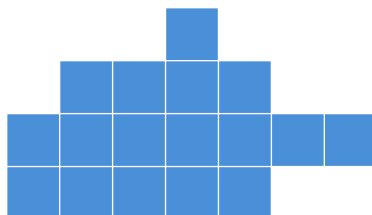
$$a(5) = 9 \text{ [PROVED] } 3 \times 5 \text{ bounding box, 9 cells.}$$



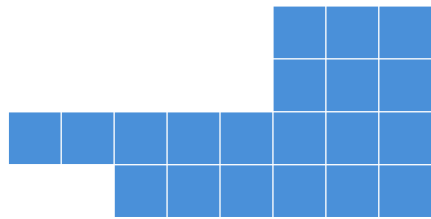
$$a(6) = 12 \text{ [PROVED] } 3 \times 6 \text{ bounding box, 12 cells.}$$



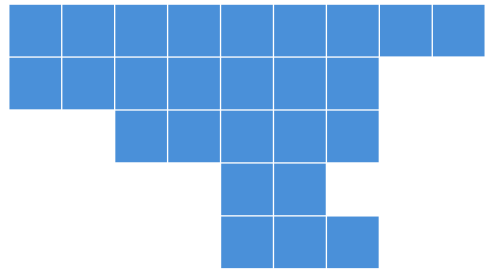
$$a(7) = 17 \text{ [PROVED] } 4 \times 7 \text{ bounding box, 17 cells.}$$



$a(8) = 20$ [PROVED] 4×8 bounding box, 20 cells.



$a(9) = 26$ [PROVED] 5×9 bounding box, 26 cells.



$a(10) = 31$ [PROVED] 5×10 bounding box, 31 cells.

